

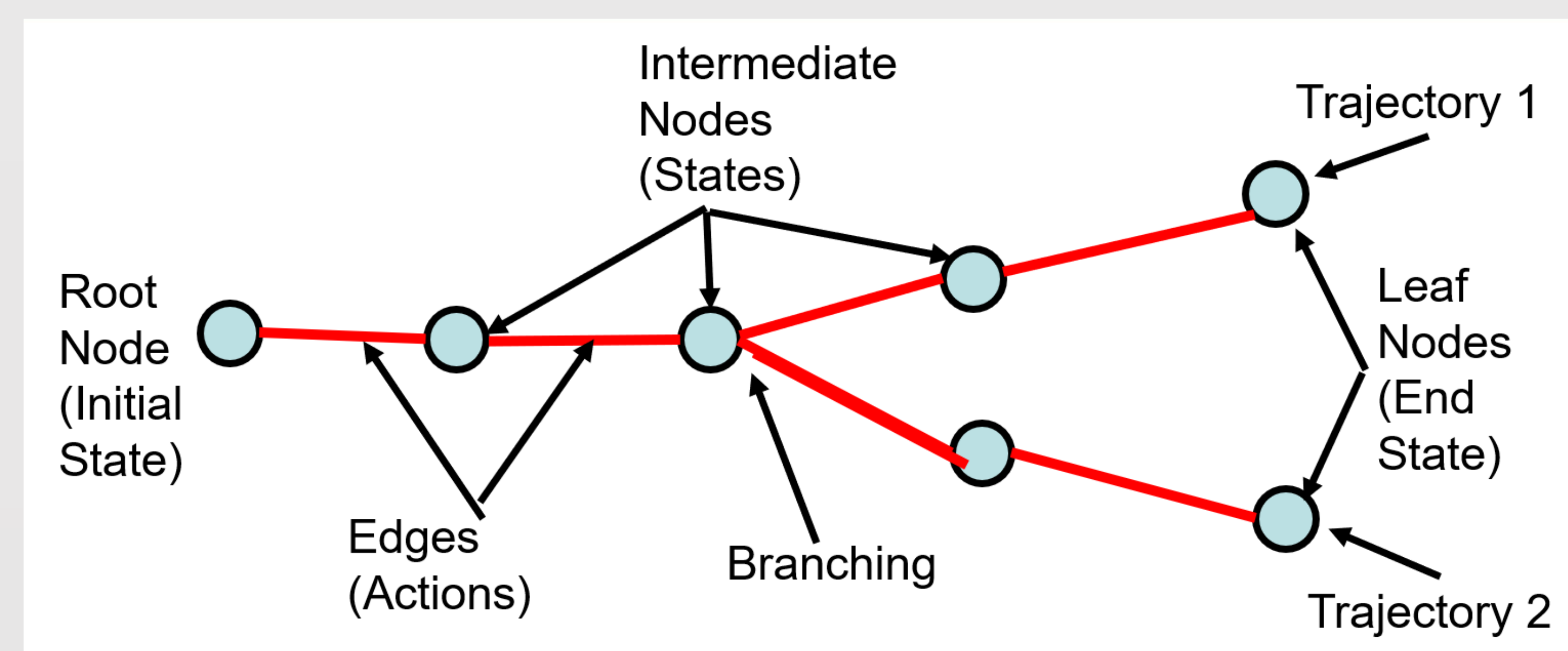
Plan Future Graph (PFG)

Dr. Bryan L. Croft, Jacob E. Viraldo, Brittany Abrams



CONCEPT

- Visualize set of AI generated plausible plans in a “Plan Future Graph” (PFG) based on a tree graph structure.
- Provide supporting user interface elements to examine and drill down into the top prioritized plans presented by the AI planner.
- Allow human planner understanding of AI generated plans (form of Explainable AI via the user interface).
- Allow human planner to assess and compare plans and then, if necessary, change parameter values to evaluate and compare sensitivity of changes.
- The PFG is still in initial design and development phase with many difficult visualization techniques observed and yet to be discovered, designed & built.



SUPPORTING INTERFACES

- The AI generated PFG represents all plausible plan outcomes and thus can be an extremely large structure involving many visualization techniques.
- A chart or map is a key component for many military planning systems and proves geo-spatial insights to the plan.
- AI driven top prioritized plan trajectory list along with corresponding highlighted trajectories and shown in the PFG structure. Drill down with corresponding UI elements provide insight to the user about the selected plan via various interface displays.
- Plan comparisons are made with parallel coordinate UI, data table, and radar chart allowing multiple views into the generated plan.
- Human can interact and modify plan resources and reset the importance of the plan criteria. The AI planning system generates a new PFG based on changes while preserving the previous version for plan comparison.



PFG TREE GRAPH

- A large tree graph structure where the root node provides a set of plans starting conditions. The root node branches out along edges representing actions of the plan taken resulting in a node representative of the next state. Leaf nodes represent ending state for a specific plan trajectory in PFG.

